

THE SHADOW OBSTRUCTION TOWER OF A CHIRAL ALGEBRA AND THE TRANSFERRED L_∞ TOWER

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ABSTRACT. Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve and let $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ be its modular convolution dg Lie algebra, with bar-intrinsic Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$. In the verified chirally Koszul genus-zero cyclic setting, after a cyclic contraction of $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$ onto its shadow cohomology is fixed, the finite-order shadow projections of $\Theta_{\mathcal{A}}$ coincide with the transferred L_∞ bracket evaluations on the projected Maurer–Cartan element. Degree two is the curvature input; degrees $r \geq 3$ are the formality obstruction evaluations. For the standard families where the transfer data are constructed, this comparison identifies the shadow termination degree with the L_∞ obstruction level.

1. INTRODUCTION

1.1. Formality and its obstructions. A dg Lie algebra $\mathfrak{g}$ is L_∞ -formal if its minimal L_∞ -model is L_∞ -isomorphic to the strict graded Lie algebra $(H^*(\mathfrak{g}), 0, [-, -]_H)$. For a fixed contraction, the transferred brackets ℓ_r^{tr} are representatives; the obstruction is their class in the appropriate deformation complex after all lower obstructions have been killed. Kontsevich’s formality theorem [3] proved that polyvector fields on a smooth manifold form an L_∞ -formal dg Lie algebra, giving deformation quantization of Poisson structures; Tamarkin [8] recovered the result operadically from formality of the little disks. When formality fails, obstructions are recorded in cohomology classes $o_3, o_4, \dots$: at degree r the class o_r measures whether ℓ_r^{tr} vanishes on cohomology; if $o_r = 0$, a homotopy kills ℓ_r^{tr} and the next obstruction lives one degree higher. The sequence $(o_r)_{r \geq 3}$ is the *formality obstruction tower*; see Loday–Vallette [5, Ch. 10].

1.2. The shadow tower of a chiral algebra. On a smooth projective curve X , a chiral algebra $\mathcal{A}$ [1] has a bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\epsilon)$ is the augmentation ideal, with deconcatenation coproduct and differential built from the Arnold form $\eta_{ij} = d \log(z_i - z_j)$ on the Fulton–MacPherson compactification of the configuration space. The modular convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is the pronilpotent dg Lie algebra whose differential encodes graph sums over stable curves in $\bar{\mathcal{M}}_{g,n}$. The bar-intrinsic element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$ satisfies $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ and packages the full genus expansion of $\mathcal{A}$ [6, MC2 and modular convolution theorem]. Its finite-degree truncations $\Theta_{\mathcal{A}}^{\leq r}$ define the *shadow obstruction tower*

$$\text{Sh}_r(\mathcal{A}) := -\nabla_H^{-1}([\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(o)}), \quad r \geq 2,$$

where the bracket is genus-zero graph composition and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\text{sh}}$. The first three shadows are $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ (the modular characteristic [6, Theorem D]), $\text{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A})$ (cubic shadow), and $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A})$ (quartic shadow).

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1.3. Shadow and transfer. The two towers are built in different complexes. The formality tower is the transferred cyclic Chevalley tower of the genus-zero convolution dg Lie algebra. The shadow tower is the scalar projection of the bar-intrinsic Maurer–Cartan recursion. The comparison is a theorem only after the genus-zero cyclic contraction is fixed.

Theorem 1.1 (Genus-zero shadow–formality comparison). *Let $\mathcal{A}$ be a chirally Koszul standard algebra for which the genus-zero cyclic convolution complex, shadow projection, and cyclic contraction (ι, π, h) are constructed. Assume the cyclic stable-tree/HPL comparison lemma: after cyclic symmetrisation, the genus-zero stable-tree graph sum and the Kadeishvili–Merkulov transfer sum have identical signs, automorphism weights, propagators, and vertex decorations. For every $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) = \ell_r^{(\circ), \mathrm{tr}}(\pi \Theta_{\mathcal{A}}^{\leq r-1}, \dots, \pi \Theta_{\mathcal{A}}^{\leq r-1}) \quad \text{in } \mathcal{A}_{r,0}^{\mathrm{sh}},$$

relative to the fixed cyclic contraction. Degree 2 is the curvature term. For $r \geq 3$ this is the transferred formality obstruction evaluated on the truncated Maurer–Cartan class.

The proof has two ingredients. The cases $r = 2, 3, 4$ are the explicit computations of Proposition 4.1. For $r \geq 5$, the Kadeishvili–Merkulov tree formula [2, 7] on the strict genus-zero convolution algebra projects, after cyclic symmetrisation, to the same stable-tree graph sum that defines the shadow coefficient.

1.4. Consequences. Theorem 1.1 identifies the shadow termination degree $r_{\max}(\mathcal{A})$ with the L_{∞} obstruction level $f_{\infty}(\mathcal{A})$ of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)}$ for the verified standard cyclic families. Operation-level equality with a transferred A_{∞} -depth requires a separate nondegeneracy statement. The four shadow-depth classes are realized by Heisenberg (class **G**, $r_{\max} = 2$, formal), affine Kac–Moody (class **L**, $r_{\max} = 3$), $\beta\gamma$ (class **C**, $r_{\max} = 4$), and Virasoro (class **M**, $r_{\max} = \infty$, intrinsically non-formal); see Section 7.

Scope. A verified chirally Koszul standard algebra is one for which the genus-zero cyclic convolution complex, the shadow cohomology projection, and a cyclic contraction (ι, π, h) have been constructed. All comparison statements below are relative to this fixed transfer datum. They do not assert formality of the full two-coloured $\mathrm{SC}^{\mathrm{ch}, \mathrm{top}}$ operad, nor of the Calabi–Yau stage-one factorisation algebra before Stage-2 specialisation.

Conventions. Grading is cohomological with $|d| = +1$; desuspension lowers degree, $|s^{-1}v| = |v| - 1$. Bar complexes use the augmentation ideal: $\tilde{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$. The propagator $d \log E(z, w)$ on the bar complex has weight one. All statements are at the level of cohomology classes unless otherwise noted.

2. THE SHADOW OBSTRUCTION TOWER

2.1. The bar complex and its convolution algebra. Let X be a smooth projective curve over $\mathbb{C}$ and let $\mathcal{A}$ be a chiral algebra on X [1]. The ordered chiral bar complex $\tilde{B}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}}) = \bigoplus_{n \geq 0} (s^{-1}\tilde{\mathcal{A}})^{\otimes n}$ carries the deconcatenation coproduct, with differential assembled from $\eta_{ij} = d \log(z_i - z_j)$ and the chiral product $\mu: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \rightarrow \Delta_* \mathcal{A}$. The *modular convolution dg Lie algebra* is

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} := \mathrm{Hom}^{\mathrm{cyc}}(\tilde{B}(\mathcal{A}), \mathcal{A}) \otimes \mathbb{G},$$

where $\mathbb{G} = \bigoplus_{g \geq 0} \mathbb{G}^{(g)}$ is the coefficient algebra generated by stable graphs on $\overline{\mathcal{M}}_{g,n}$. The genus filtration gives $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \bigoplus_{g \geq 0} (\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})_g$; the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, (\circ)} = \mathrm{Hom}^{\mathrm{cyc}}(\tilde{B}(\mathcal{A}), \mathcal{A}) \otimes \mathbb{G}^{(\circ)}$ is a strict dg Lie algebra with bracket $[-, -]^{(\circ)}$ from tree composition, since genus-zero stable graphs are trees.

2.2. The Maurer–Cartan element and degree truncations. The differential on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ splits as $D = d_o + D_{\text{int}}$, with d_o linear and D_{int} the interacting piece encoding the chiral product and the Arnold form. The difference $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o \in \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is a Maurer–Cartan element [6, MC2 bar-intrinsic construction]:

$$D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The element $\Theta_{\mathcal{A}}$ is bar-intrinsic. Write

$$F_{\leq r}\bar{B}(\mathcal{A}) = \bigoplus_{n \leq r} (s^{-1}\bar{\mathcal{A}})^{\otimes n}, \quad F_{> r}\bar{B}(\mathcal{A}) = \bigoplus_{n > r} (s^{-1}\bar{\mathcal{A}})^{\otimes n}.$$

The truncation $\Theta_{\mathcal{A}}^{\leq r}$ is the projection of $\Theta_{\mathcal{A}}$ modulo $F_{> r}$. The Maurer–Cartan recursion determines each new degree up to a coboundary from the previous truncation.

Definition 2.1 (Shadow obstruction tower). The *degree- r shadow* of $\mathcal{A}$ is

$$\text{Sh}_r(\mathcal{A}) := -\nabla_H^{-1}([\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(o)}) \in \mathcal{A}_{r,o}^{\text{sh}},$$

where the bracket is the genus-zero graph composition, the subscript r extracts the degree- r component, and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\text{sh}} = H^*(\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)})$ descended from the convolution Lie bracket. The sequence $(\text{Sh}_r(\mathcal{A}))_{r \geq 2}$ is the *shadow obstruction tower* of $\mathcal{A}$.

The definition is bar-intrinsic: if $\Theta_{\mathcal{A}} \equiv \Theta'_{\mathcal{A}} \text{ mod gauge}$, then $\text{Sh}_r(\mathcal{A})$ and $\text{Sh}'_r(\mathcal{A})$ coincide as cohomology classes in $\mathcal{A}_{r,o}^{\text{sh}}$.

Remark 2.2 (Closed formulas). On the scalar projection, writing $\Theta_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$ and $H(t) = \sum_{r \geq 2} r S_r t^r$, one has $\text{Sh}_2 = \kappa$, $\text{Sh}_3 = \mathfrak{C}$, $\text{Sh}_4 = \mathfrak{Q}$, and the shadow metric identity

$$H(t)^2 = t^4(4\kappa^2 + 12\kappa\mathfrak{C}t + (9\mathfrak{C}^2 + 16\kappa\mathfrak{Q})t^2)$$

on the families where the Riccati reduction has been proved.

3. THE L_{∞} -FORMALITY OBSTRUCTION TOWER

3.1. Homotopy transfer on a strict model. Let $\mathfrak{g}$ be a strict dg Lie algebra with differential d and bracket $[-, -]$. A *deformation retract* onto $H^*(\mathfrak{g})$ is a triple (ι, π, h) with $\iota: H^*(\mathfrak{g}) \rightarrow \mathfrak{g}$, $\pi: \mathfrak{g} \rightarrow H^*(\mathfrak{g})$, and $h: \mathfrak{g} \rightarrow \mathfrak{g}$ of degree -1 , satisfying $\pi\iota = \text{id}$ and $\text{id} - \iota\pi = [d, h]$. The homotopy transfer theorem of Kadeishvili [2] and Kontsevich–Soibelman [4], in the form of Loday–Vallette [5, §10.3], produces a transferred L_{∞} -algebra structure $(H^*(\mathfrak{g}), o, \{\ell_n^{\text{tr}}\}_{n \geq 2})$ quasi-isomorphic to $\mathfrak{g}$, with brackets computed by the Kadeishvili–Merkulov tree sum

$$\ell_r^{\text{tr}}(a_1, \dots, a_r) = \sum_{T \in \text{Trees}_r} \frac{\pm 1}{|\text{Aut}(T)|} \pi \circ \Phi_T(\iota(a_1), \dots, \iota(a_r)), \quad (3.1)$$

where Trees_r is the set of planar rooted trees with r leaves and Φ_T decorates internal vertices with $[-, -]$ and internal edges with h .

3.2. The formality obstruction tower.

Definition 3.1 (L_{∞} -formality and obstructions). The dg Lie algebra $\mathfrak{g}$ is *L_{∞} -formal* if there is an L_{∞} quasi-isomorphism $\mathfrak{g} \simeq (H^*(\mathfrak{g}), o, [-, -])$; equivalently, $\ell_n^{\text{tr}} = 0$ in L_{∞} cohomology for all $n \geq 3$. The *L_{∞} -formality obstruction tower* is the sequence

$$(o_r(\mathfrak{g}))_{r \geq 3}, \quad o_r(\mathfrak{g}) := [\ell_r^{\text{tr}}] \in H^*(\text{CE}_{\text{cyc}}^{\bullet}(\mathfrak{g})).$$

If $o_r(\mathfrak{g}) = 0$, an L_{∞} homotopy kills ℓ_r^{tr} and the next obstruction is defined.

For a pronilpotent dg Lie algebra with Maurer–Cartan element Θ , evaluating the transferred brackets on Θ produces a refined tower $(\ell_r^{\text{tr}}(\Theta, \dots, \Theta))_{r \geq 2}$, which takes values in the convolution target. Obstruction cohomology records the projected class. The refined target-valued tower is the one compared with the shadow tower.

Remark 3.2 (Cyclic deformation complex). For $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$, the relevant deformation complex is the cyclic Chevalley complex of the genus-zero convolution dg Lie algebra. Under the chirally Koszul hypotheses and the chosen cyclic contraction, this complex maps to the shadow cohomology $\mathcal{A}_{*,o}^{\text{sh}}$ by the projection π .

4. BASE CASES: DEGREES 2, 3, 4

Proposition 4.1 (Shadow–formality identification at low degree). *For every verified chirally Koszul standard algebra in the fixed genus-zero cyclic transfer setting, the following identities hold:*

- (i) $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A}) = \ell_2^{(o)}(\pi\Theta_{\mathcal{A}}, \pi\Theta_{\mathcal{A}}).$
- (ii) $\text{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(o)}(\pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2})).$
- (iii) $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(o),\text{tr}}(\pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3})].$

Proof. *Part (i).* The binary bracket on $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$ is the genus-zero convolution Lie bracket; evaluating on $\Theta_{\mathcal{A}}$ produces the scalar curvature $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ by definition [6, Theorem D]. The identity $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ is the degree-2 scalar shadow, so the three quantities agree.

Part (ii). The ternary bracket $\ell_3^{(o)}$, computed by (3.1), projects after cyclic symmetrisation to the three s -, t -, and u -channel stable four-point trees. On the primary line $[\ell_3^{(o)}] = 0$ since $H^1(\overline{\mathcal{M}}_{0,4}) = H^1(\mathbb{P}^1) = 0$; at the chain level $\ell_3^{(o)}$ is exact and nonzero, and homotopy transfer gives $\mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(o)}(\pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2})).$ The cubic shadow $\mathfrak{C}(\mathcal{A})$, computed as the three-channel graph sum at degree three, matches the cyclic projection of the transferred tree sum with identical weights and propagators.

Part (iii). The quaternary bracket $\ell_4^{(o)}$ is a planar HPL tree sum whose cyclic projection is indexed by the 15 labelled genus-zero stable trees with five external legs. Its transferred class $\ell_4^{(o),\text{tr}}$ lies in $H^*(\overline{\mathcal{M}}_{0,5})$. The weight-four graph sum defining $\mathfrak{Q}(\mathcal{A})$ enumerates those labelled stable trees with matching weights, signs, and propagators [6, appendix on nonlinear modular shadows], so $\text{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(o),\text{tr}}(\pi\Theta_{\mathcal{A}}^{\leq 3}, \dots, \pi\Theta_{\mathcal{A}}^{\leq 3})].$ $\square$

Remark 4.2 (Degree-2: κ is family-dependent). The identity $\text{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ holds at degree two on the scalar projection for the verified standard families; on the primary line this is the definition $S_2 := \kappa$. What is family-dependent is the value of κ itself: $\kappa^{\text{Vir}} = c/2$, $\kappa^{\text{Heis}} = k$, $\kappa^{\text{KM}} = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$, $\kappa^{\text{WN}} = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$. The formula $\kappa = c/2$ is Virasoro-specific.

5. THE COMPARISON PROOF

Theorem 5.1 (Genus-zero shadow–formality comparison). *Let $\mathcal{A}$ be a verified chirally Koszul standard algebra in the genus-zero cyclic setting, with fixed cyclic contraction (ι, π, h) of $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$ onto $\mathcal{A}_{*,o}^{\text{sh}}$. Assume the cyclic stable-tree/HPL comparison lemma. For every degree $r \geq 2$,*

$$\text{Sh}_r(\mathcal{A}) = \ell_r^{(o),\text{tr}}(\pi\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \pi\Theta_{\mathcal{A}}^{\leq r-1}) \quad \text{in } \mathcal{A}_{r,o}^{\text{sh}}, \quad (5.1)$$

where $\ell_r^{(o),\text{tr}}$ is the transferred genus-zero L_∞ bracket on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(o)}$. For $r = 2$ this is the curvature input. For $r \geq 3$ it is the formality obstruction evaluation on the truncated Maurer–Cartan element.

Proof. Since genus-zero stable graphs are trees, $\mathfrak{g}_{\mathcal{A}}^{\text{mod},(0)}$ is a strict dg Lie algebra. The fixed cyclic contraction satisfies $\pi\iota = \text{id}$, $\text{id} - \iota\pi = [D, h]$. The homotopy transfer theorem [2, 4, 5] produces a transferred L_∞ structure $(\mathcal{A}_{*,0}^{\text{sh}}, \{\ell_r^{(0),\text{tr}}\})$, with brackets computed by (3.1).

At $r = 2$, both sides equal $\kappa(\mathcal{A})$ by Proposition 4.1(i); the cases $r = 3, 4$ are parts (ii), (iii). For $r \geq 5$, assume (5.1) through degree $r - 1$. The cyclic stable-tree/HPL comparison lemma identifies the graph sum for $\text{Sh}_r(\mathcal{A})$ with the Kadeishvili–Merkulov tree formula for $\ell_r^{(0),\text{tr}}(\pi\Theta^{\leq r-1}, \dots, \pi\Theta^{\leq r-1})$: internal edges carry h , outputs are projected by π , and internal vertices are the lower-degree transferred brackets. The projected sums agree in $\mathcal{A}_{*,0}^{\text{sh}}$. $\square$

Remark 5.2 (Genus-zero projection and positive-genus terms). Theorem 5.1 concerns the genus-zero projection. For scalar uniform-weight families the full shadow tower has the expansion

$$\text{Sh}_r(\mathcal{A}) = \underbrace{\ell_r^{(0),\text{tr}}(\pi\Theta^{\leq r-1}, \dots)}_{\text{genus-zero formality}} + \underbrace{\sum_{g \geq 1} \ell_r^{(g),\text{tr}}(\dots) + \Lambda_P(\dots)}_{\text{genus } \geq 1 \text{ corrections}},$$

where Λ_P is the BV loop operator. The first summand is controlled by classical homotopy transfer. The positive-genus summands are governed by the quantum L_∞ operations and the BV loop operator. At genus $g \geq 2$ with multi-weight generators, the scalar formula for the genus corrections receives a cross-channel term $\partial F_g^{\text{cross}}$; the genus-zero comparison (5.1) is unchanged.

6. COMMON DEPTH

Definition 6.1 (Two genus-zero depth invariants). For a chiral algebra $\mathcal{A}$, set $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : \text{Sh}_r(\mathcal{A}) \neq 0\}$ (shadow termination degree), and $f_\infty^\Theta(\mathcal{A}) := \sup\{n \geq 2 : \ell_n^{(0),\text{tr}}(\pi\Theta^{\leq n-1}, \dots, \pi\Theta^{\leq n-1}) \neq 0\}$ (evaluated L_∞ obstruction level on the transferred genus-zero model).

Corollary 6.2 (Evaluated common depth for the standard cyclic families). *For the verified chirally Koszul standard families in the genus-zero cyclic setting,*

$$r_{\max}(\mathcal{A}) = f_\infty^\Theta(\mathcal{A}).$$

Proof. The equality is the evaluation identity of Theorem 5.1, degree by degree. Operation-level vanishing requires an independent nondegeneracy lemma for the transferred cyclic primitive component. $\square$

Corollary 6.3 (Coarse depth realisation). *Within the constructed standard cyclic families, the four coarse shadow-depth classes are all realized: family:*

$$\mathbf{G}(r_{\max} = 2), \quad \mathbf{L}(r_{\max} = 3), \quad \mathbf{C}(r_{\max} = 4), \quad \mathbf{M}(r_{\max} = \infty).$$

The class $\mathbf{G}$ consists of L_∞ -formal algebras; the class $\mathbf{M}$ consists of intrinsically non-formal algebras, in the sense that no finite truncation of their minimal L_∞ model is quasi-isomorphic to a dg Lie algebra.

Proof. Theorem 5.1 identifies shadow depth with the L_∞ obstruction level after evaluation on the transferred Maurer–Cartan element. The examples in Section 7 realize each coarse class. $\square$

7. EXAMPLES

The four standard families realize the four shadow-depth classes.

Example 7.1 (Heisenberg; class $\mathbf{G}$, formal). The Heisenberg vertex algebra Heis_k at level k , with $J(z)J(w) \sim k/(z-w)^2$, has $\kappa(\text{Heis}_k) = k$. The bar differential is strictly commutative, so $m_n = 0$ and $\ell_n^{(0),\text{tr}} = 0$ for $n \geq 3$. Hence $r_{\max}(\text{Heis}_k) = 2$ and Heis_k is L_∞ -formal. Class $\mathbf{G}$.

Example 7.2 (Affine Kac–Moody; class **L**). At non-critical level $k \neq -h^\vee$, the affine Kac–Moody vertex algebra $\widehat{\mathfrak{g}}_k$ has $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. The binary bracket is the Lie bracket on $\mathfrak{g}$; Jacobi kills all degree-four graph contributions, so $\ell_4^{(o),\text{tr}} = 0$, while $\ell_3^{(o),\text{tr}} \neq 0$ in general. Hence $r_{\max}(\widehat{\mathfrak{g}}_k) = 3$. Class **L**.

Example 7.3 ($\beta\gamma_\lambda$; class **C**, shadow depth 4). The standard rank-one $\beta\gamma_\lambda$ contact family has

$$S_2 = 6\lambda^2 - 6\lambda + 1, \quad S_3 = 0, \quad S_4 = -\frac{5}{12}, \quad S_r = 0 \quad (r \geq 5).$$

Thus $r_{\max}(\beta\gamma_\lambda) = 4$ at the generic contact points. Equivalently, the bracket $\ell_3^{(o),\text{tr}}$ vanishes and $\ell_4^{(o),\text{tr}}$ is nonzero. Class **C**.

Example 7.4 (Virasoro; class **M**, obstructed at every degree). The Virasoro vertex algebra Vir_c has $\kappa(\text{Vir}_c) = c/2$. The transferred brackets on the primary line of $H^*(\tilde{B}(\text{Vir}_c))$ satisfy $m_k^{\text{tr}}(s^{-1}T, \dots, s^{-1}T) = S_k \cdot e_{2k}$ for $k \geq 2$, with $s^{-1}T$ the desuspension of the conformal vector to bar degree one and e_{2k} the weight- $2k$ basis vector of Ext^1 . The first shadow coefficients are

$$\begin{aligned} S_2 &= \frac{c}{2}, & S_3 &= 2 \text{ (} c\text{-independent)}, & S_4 &= \frac{10}{c(5c+22)}, \\ S_5 &= -\frac{48}{c^2(5c+22)}, & S_6 &= \frac{4(240c+1031)}{c^3(5c+22)^2}. \end{aligned}$$

The weighted Ricci-metric coefficients $\widehat{S}_6^{wt}$ and $\widehat{S}_7^{wt}$ use a different normalization. Here Ext^1 denotes the primary bar-cohomology line in the transferred shadow model. It is not a Hochschild-degree statement; Theorem H is the separate chiral Hochschild concentration theorem in degrees $\{0, 1, 2\}$, with symmetric descent obtained by averaging. For generic c on the nonsingular Virasoro shadow surface, the Ricci recurrence and computed coefficients give infinitely many nonzero S_k . Thus $r_{\max}(\text{Vir}_c) = \infty$. The normal-ordered composites are the source of possible operations, not the nonvanishing proof. Class **M**.

Remark 7.5 (The four standard depths). For the verified finite-type standard families governed by the quadratic Ricci reduction, the coarse shadow-depth classes are **G**, **L**, **C**, and **M**. The identity $H(t)^2 = t^4 Q(t)$ with $\deg Q \leq 2$ gives depths 2, 3, 4, and ∞ for this scalar projection [6, theorem on Ricci algebraicity]. This is a coarse classification of the standard cyclic families. Fine finite-depth towers require separate analysis.

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